

Hypothesis Testing

Hypothesis testing is a method used to decide whether sample data provides enough evidence to support or reject a claim about a population.

Think of it as a formal way of answering:

"Is this observed result just random chance, or is something real happening?"

Example: Battery Life Claim

Suppose a company claims:

"Our batteries last 10 hours on average."

You test 100 batteries and find:

- Sample mean = 9.2 hours

Now you want to know:

Is the difference (9.2 vs 10) large enough to conclude the claim is false?

This is a hypothesis testing problem.

Step 1: State the Hypotheses

Null Hypothesis (H_0)

The default assumption.

$H_0: \mu = 10$

Meaning:

The average battery life is 10 hours.

Alternative Hypothesis (H_1)

The claim you're investigating.

$H_1: \mu \neq 10$

Meaning:

The average battery life is not 10 hours.

Step 2: Assume the Null Hypothesis Is True

This is important.

We start by pretending the company's claim is correct.

Then we ask:

If the true mean really is 10 hours, how likely is it to observe a sample mean as extreme as 9.2?

Step 3: Calculate a Test Statistic

The most common is the z-statistic:

$$z = \frac{\bar{x} - \mu}{SE}$$

where:

- $\bar{x}$ = sample mean
- μ = hypothesized mean
- SE = standard error

The z-score measures how many standard errors the sample mean is away from the claimed mean.

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$z = \frac{(1.2) - (0)}{1} = 1.2$$

$$\Phi(z) \approx 88.5\%$$

xxx

1.

μ

0.

σ

1.

Step 4: Find the p-value

The **p-value** answers:

Assuming H_0 is true, what is the probability of obtaining a result at least this extreme?

Small p-value:

- Result is unlikely under H_0
- Evidence against H_0

Large p-value:

- Result is plausible under H_0
- Not enough evidence against H_0

Step 5: Compare With Significance Level

Usually:

$\alpha = 0.05$

(5% significance level)

Decision rule:

- If $p < 0.05$, reject H_0
- If $p \geq 0.05$, fail to reject H_0

Example

Suppose:

- Claimed mean = 10
- Sample mean = 9.2
- Standard error = 0.25

Then:

$$z = \frac{9.2 - 10}{0.25} = -3.2$$

A z-score of -3.2 is quite far from 0.

The corresponding p-value is approximately:

$$p \approx 0.0014$$

Since:

$$0.0014 < 0.05$$

we reject H_0 .

Conclusion:

There is strong evidence that the average battery life is not 10 hours.

Connection to Confidence Intervals

Confidence intervals and hypothesis tests are closely related.

Suppose a 95% confidence interval is:

$$(68.04, 71.96)$$

If someone claims:

$$\mu = 75$$

Since 75 is outside the interval:

- Reject $H_0: \mu = 75$
- at the 5% significance level

If the claimed value is inside the interval:

- Fail to reject H_0

Possible Outcomes

Correct Decisions

Reality	Decision
H_0 true	Fail to reject
H_0 false	Reject

Errors

Type I Error

Reject a true null hypothesis.

Example:

- Drug actually works.
- Study incorrectly concludes it doesn't.

Probability:

$P(\text{Type I Error}) = \alpha$

Usually 5%.

Type II Error

Fail to reject a false null hypothesis.

Example:

- Drug doesn't work.
- Study concludes there's insufficient evidence against it.

Real-Life Examples

Medical Research

Claim:

New drug lowers blood pressure.

Hypothesis test determines whether observed improvement is statistically significant.

A/B Testing

Claim:

New website design increases conversions.

Hypothesis test compares conversion rates.

Manufacturing

Claim:

Average product weight is 500 g.

Samples are tested to ensure quality standards.

Election Polls

Claim:

Candidate A is leading Candidate B.

Hypothesis testing helps determine whether the observed lead is likely real or due to sampling variation.

The Big Picture

Hypothesis testing is essentially:

1. Make a claim (H_0)

2. Assume it's true
 3. Measure how unusual your sample is under that assumption
 4. Use probability to decide whether the sample is too unusual to be explained by chance
- That's why probability is the foundation of inferential statistics: hypothesis testing is fundamentally a probability-based decision process.